

Step 1: Data

Before starting to train a model, please get familiar with the dataset. When you look at the dataset, please answer the following questions:

- 1) How many data samples are included in the dataset?
For training there are 79 image-mask pairs
For testing there are 19 image-mask pairs
- 2) Which problem will this dataset try to address?
We are trying to achieve accurate segmentation of blood vessels from the retina blood vessels segmentation dataset. This technique can be used for medical image analysis and enhancing the diagnosis of retinal vascular diseases.
- 3) What is the dimension ranging in the dataset?
The dimension of all images is (512, 512, 3)
- 4) Does the dataset have any missing information? E.g., missing features.
The dataset does not have any missing information
- 5) What is the label of this dataset?
In this dataset the label is its corresponding segmentation masks. The mask is the ground truth. The mask is a binary image where each value is either 0 or 1
- 6) How many percent of data will you use for training, validation and testing?
I will be splitting the training data set as 80% training and 20% validation. Also, I will be using entire test dataset for testing
- 7) What kind of data pre-processing will you use for your training dataset?
Firstly, I converted all loaded images to RGB format. Then I normalized images and masks to range [0,1] by dividing the pixel values by 255. I also threshold the mask values at 0.5 to make it binary segmentation.

Step 2: Model

- Here I selected U-Net model, you will experience different hyperparameters (including model architecture). Please change the following hyperparameters and report the model performance in the testing dataset

I created two models: UNet with 3 layers and UNet with 2 layers. At start I kept facing plateau results, so I switched to a scheduler which starts with low learning rate and gradually increases it.

Step 3: Objective

Medical Image Segmentation

- Which loss function are you going to use to train your model?

I used combination of focal loss and Dice loss. Dice loss was important to find the overlapping between the prediction and the actual image. Focal loss was used to help with the imbalanced dataset. Without the focal loss the model kept predicting the background/black color, lowering the model IoU and Dice values. I tried different weights and found out the 70% Dice Loss and 30% Focal loss provided the best model performance.

Step 4: Optimization

- You are going to select your optimization function to train the models. Report the optimization you selected in this section and explain the reason for using the optimizer.

I used Adam optimizer for this problem. Adam optimizer is preferred for CNN related tasks as it usually outperforms other optimizers.

Step 5: Model selection

- Based on your training experience, which model gives the best performance (accuracy). Have you experienced different learning rates?

Model	Dice and IoU with normalization	Dice and IoU without normalization
U-Net 2 blocks	IoU = 0.59 Dice = 0.74	Iou = 0.55 Dice = 0.69
U-Net 3 blocks	IoU = 0.65 Dice = 0.75	Iou = 0.53 Dice = 0.68

- How do you avoid overfitting your model and underfit your model?

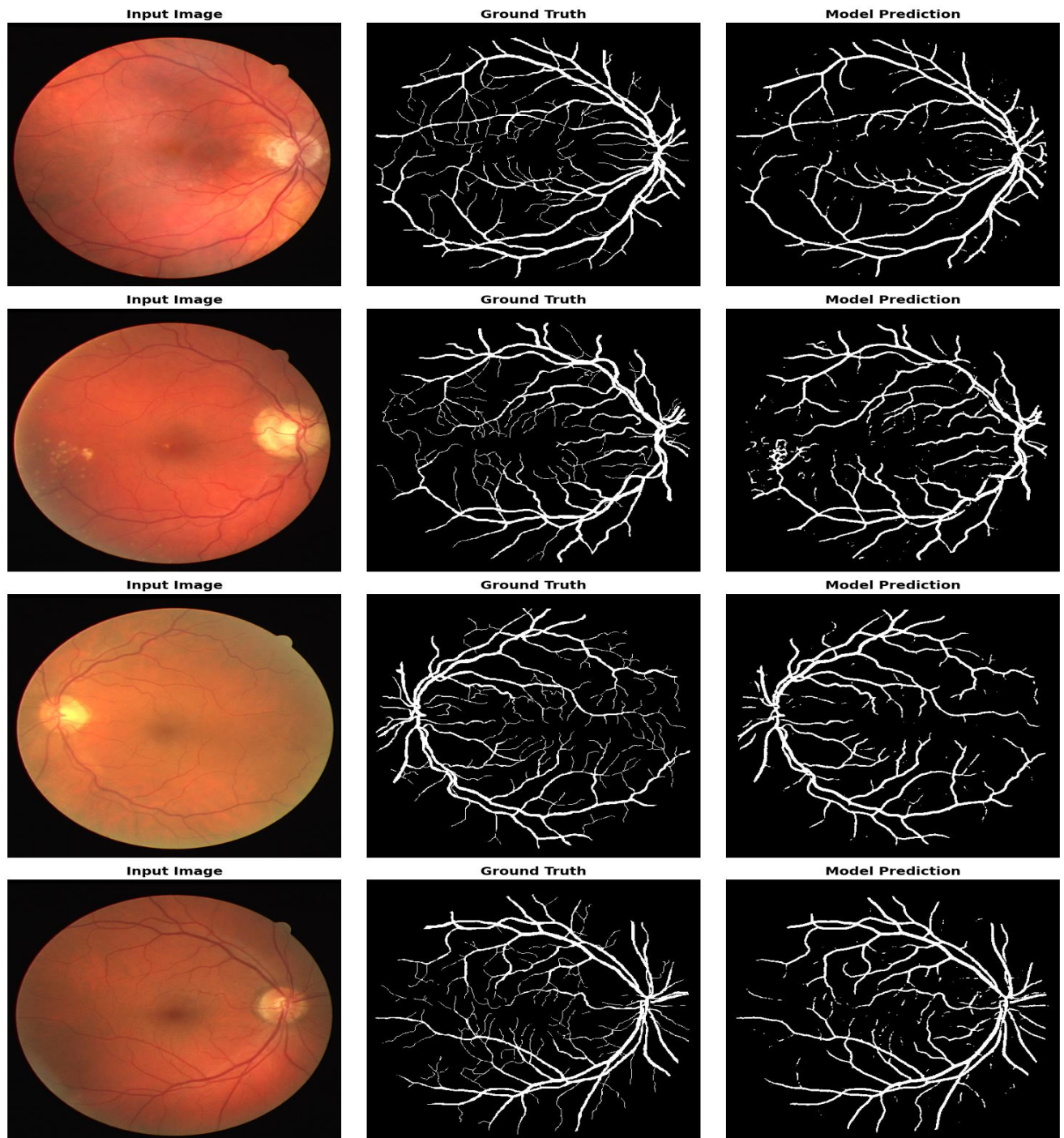
I used learning rate scheduler and Combined Loss Function of Focal and Dice.

Step 6: Model demo

Medical Image Segmentation

- In this step you should report your model prediction demo, which you did in the previous steps. Show 4 text images with label and models prediction as example showing below:

U-Net 2 blocks



U-Net 3 blocks

Medical Image Segmentation

